Q1 - 25 July - Shift 1

The area of the region given by

$A=\{(x, y) : x^2 \leq y \leq \min \{x + 2, 4 - 3x\}\}$ is :

(A) $\frac{31}{8}$ (B) $\frac{17}{6}$ (C) $\frac{19}{6}$ (D) $\frac{27}{8}$

Space for your notes:

Q2 - 26 July - Shift 1

The odd natural number a, such that the area of the region bounded by $y = 1$, $y = 3$, $x = 0$, $x = y^a$ is $\frac{364}{3}$, equal to :

(A) 3 (B) 5

(C) 7 (D) 9

Space for your notes:

Q3 - 26 July - Shift 2

The area bounded by the curves $y = |x^2-1|$ and $y = 1$ is

(A) $\frac{2}{3}(\sqrt{2}+1)$ (B) $\frac{4}{3}(\sqrt{2}-1)$

(C) $2(\sqrt{2}-1)$ (D) $\frac{8}{3}(\sqrt{2}-1)$

Space for your notes:

Q4 - 27 July - Shift 1

The area of the smaller region enclosed by the curves $y^2 = 8x + 4$ and $x^2 + y^2 + 4\sqrt{3}x - 4 = 0$ is equal to

(A) $\frac{1}{3}\left(2 - 12\sqrt{3} + 8\pi\right)$

(B) $\frac{1}{3}\left(2 - 12\sqrt{3} + 6\pi\right)$

(C) $\frac{1}{3}\left(4 - 12\sqrt{3} + 8\pi\right)$

(D) $\frac{1}{3}\left(4 - 12\sqrt{3} + 6\pi\right)$

Space for your notes:

Q5 - 27 July - Shift 2

The area of the region enclosed by

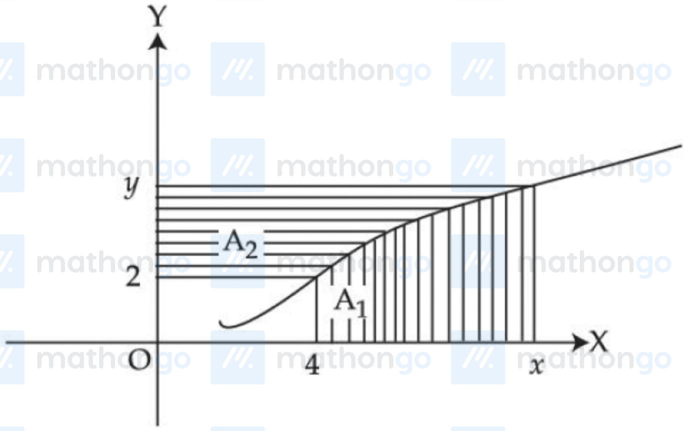
$y \le 4x^2, x^2 \le 9y$ and $y \le 4$, is equal to :

(A) $\frac{40}{3}$ (B) $\frac{56}{3}$ (C) $\frac{112}{3}$ (D) $\frac{80}{3}$

Space for your notes:

Q6 - 27 July - Shift 2

Consider a curve $y = y(x)$ in the first quadrant as shown in the figure. Let the area A_1 is twice the area A_2. Then the normal to the curve perpendicular to the line $2x - 12y = 15$ does **NOT** pass through the point.



(1) (6, 21) (2) (8, 9)
(3) (10, –4) (4) (12, –15)

Space for your notes:

Q7 - 28 July - Shift 2

The area enclosed by the curves $y = \log_e (x + e^2)$, $x = \log_e\left(\frac{2}{y}\right)$ and $x = \log_e 2$, above the line $y = 1$ is

(A) $2+e-\log_e 2$ (B) $1+e-\log_e 2$
(C) $e-\log_e 2$ (D) $1+\log_e 2$

Space for your notes:

Q8 - 29 July - Shift 1

The area of the region $\left\{(x, y) : |x-1| \leq y \leq \sqrt{5-x^2}\right\}$ is equal to :

(A) $\frac{5}{2}\sin^{-1}\left(\frac{3}{5}\right)-\frac{1}{2}$ (B) $\frac{5\pi}{4}-\frac{3}{2}$

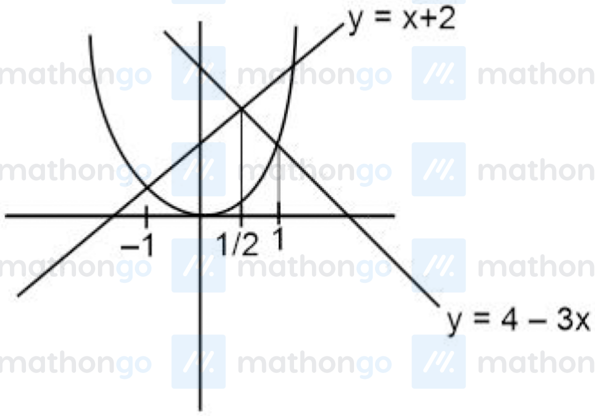
(C) $\frac{3\pi}{4}+\frac{3}{2}$ (D) $\frac{5\pi}{4}-\frac{1}{2}$

Space for your notes:

Answer Key

Q1 (B)	**Q2** (B)	**Q3** (D)	**Q4** (C)
Q5 (D)	**Q6** (C)	**Q7** (B)	**Q8** (D)

Q1 (B)



$$A=\int_{-1}^{\frac{1}{2}}\left(x+2-x^2\right)dx+\int_{\frac{1}{2}}^{1}\left(4-3x-x^2\right)dx=\frac{17}{6}$$

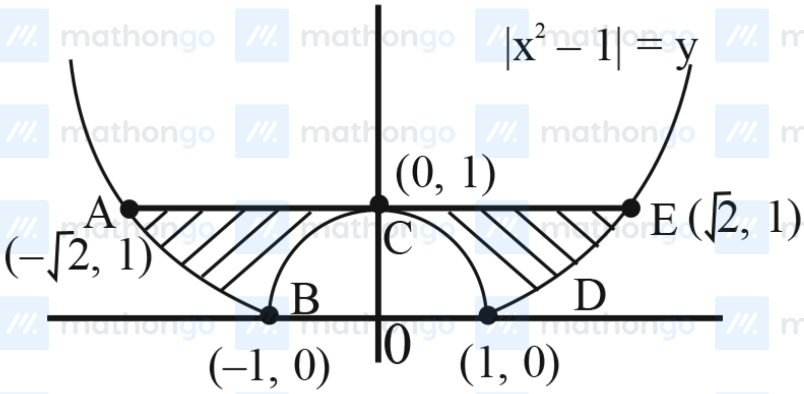
Q2 (B)

$$A=\int_{1}^{3}y^{a}.dy=\left.\frac{y^{a+1}}{a+1}\right|_{1}^{3}=\frac{364}{3}$$

$$\Rightarrow a=5$$

Q3 (D)

$y = |x^2 - 1|$

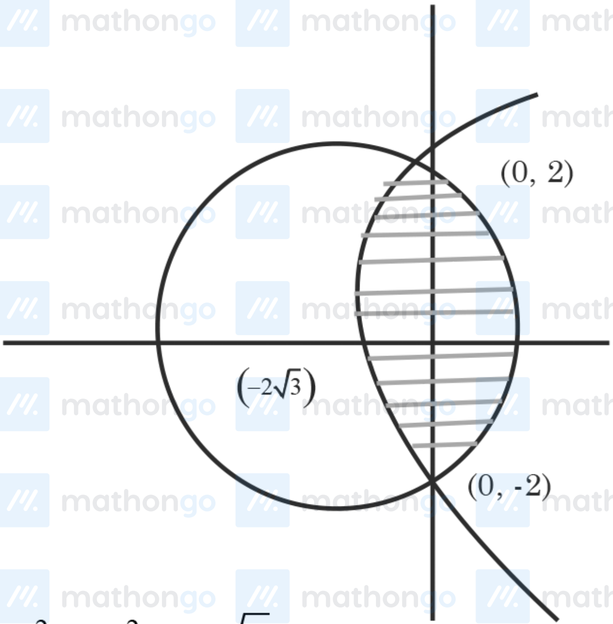


Area = ABCDEA

$$= 2\left(\int_0^1 \left(1-(1-x^2)\right)dx + \int_1^{\sqrt{2}} \left(1-(x^2-1)\right)dx\right)$$

$$= \frac{8}{3}(\sqrt{2}-1)$$

Q4 (C)



$x^2 + y^2 + 4\sqrt{3}x - 4 = 0$

$y^2 = 8x + 4$

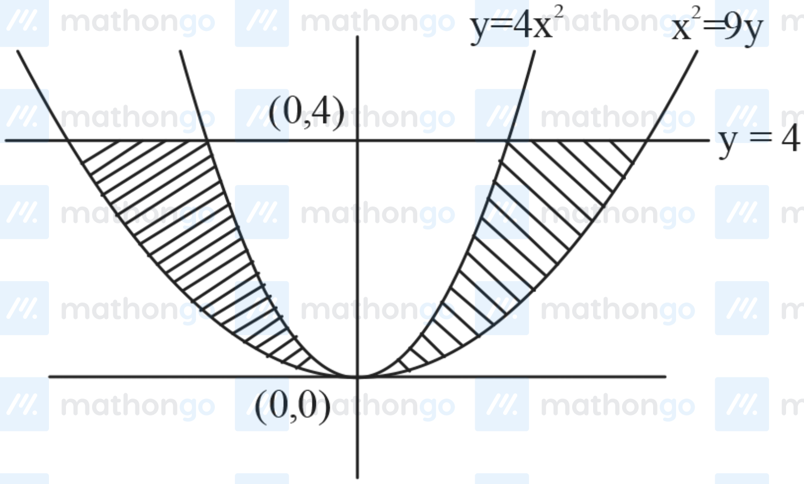
Point of intersections are (0, 2) & (0, -2)

Both are symmetric about x-axis

$$\text{Area } = 2\int_0^2 \left(\sqrt{16-y^2} - 2\sqrt{3}\right) - \left(\frac{y^2-4}{8}\right) \text{dy}$$

On solving $Area = \frac{1}{3}\left[8\pi + 4 - 12\sqrt{3}\right]$

Q5 (D)



$$\Delta = 2 \cdot \int_0^4 \left(3\sqrt{y} - \frac{\sqrt{y}}{2}\right) dy$$

$$= \quad 2 \cdot \int_0^4 \frac{5}{2}\sqrt{y}\, dy = \frac{80}{3}$$

Q6 (C)

Given that $A_1 = 2A_2$

from the graph $A_1 + A_2 = xy - 8$

$$\Rightarrow \frac{3}{2}A_1 = xy - 8$$

$$\Rightarrow A_1 = \frac{2}{3}xy - \frac{16}{3}$$

$$\Rightarrow \int_4^x f(x)dx = \frac{2}{3}xy - \frac{16}{3}$$

$$\Rightarrow f(x) = \frac{2}{3}\left(x\frac{dy}{dx} + y\right)$$

$$\Rightarrow \frac{2}{3}x\frac{dy}{dx} = \frac{y}{3}$$

$$\Rightarrow 2\int\frac{dy}{y} = \int\frac{dx}{x}$$

$$\Rightarrow 2\ell ny = \ell nx + \ell nc$$

$$\Rightarrow y^2 = cx$$

As $f(4) = 2 \Rightarrow c = 1$

so $y^2 = x$

slope of normal $= -6$

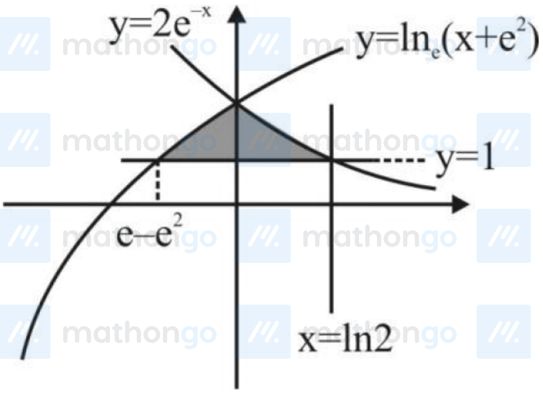
$$y = -6(x) - \frac{1}{2}(-6) - \frac{1}{4}(-6)^3$$

$$\Rightarrow y = -6x + 3 + 54$$

$$\Rightarrow y + 6x = 57$$

Now check options and (C) will not satisfy.

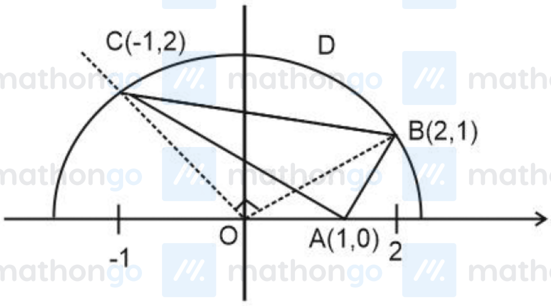
Q7 (B)



Required area is

$$= \int_{e-e^2}^{0} \ell n\left(x+e^2\right)-1dx+\int_{0}^{\ell n2} 2e^{-x}-1dx = 1+e-\ell n2$$

Q8 (D)



$|x-1|<y<\sqrt{5-x^2}$

When $|x-1|=\sqrt{5-x^2}$

$\Rightarrow (x-1)^2=5-x^2$

$\Rightarrow x^2-x-2=0$

$\Rightarrow x=2,-1$

Required Area = Area of ΔABC + Area of region BCD

$$=\frac{1}{2}\begin{vmatrix}1 & 0 & 1\\ 2 & 1 & 1\\ -1 & 2 & 1\end{vmatrix}+\frac{\pi}{4}\left(\sqrt{5}\right)^2-\frac{1}{2}\left(\sqrt{5}\right)^2$$

$$=\frac{5\pi}{4}-\frac{1}{2}$$